

SM

Model Report

Generated by bsm_agent — Shaikh Saad

Contents

1	Field Content and Anomalies	1
1.1	Standard Model Field Content	1
1.2	Anomaly Summary	1
2	Compact Form of the Full Lagrangian	1
2.1	Yukawa and Fermion-Mass Terms	1
2.2	Scalar Potential	2
2.3	Gauge and Kinetic Terms	2
3	Fully Expanded Form of the Lagrangian	2
3.1	Yukawa and Fermion-Mass Terms	2
3.2	Scalar Potential	2
3.3	Gauge and Kinetic Terms	2
4	EWSB Mass Matrices	14
4.1	VEV Substitutions	14
4.2	Stationary Condition	14
4.3	Neutral Scalar Mass Matrix	15
4.4	Scalar Hermitian Blocks	15
4.5	Scalar Holomorphic Blocks	15
4.6	Fermion Blocks	15

1 Field Content and Anomalies

1.1 Standard Model Field Content

q	=	weyl fermion	$(3, 2, 1/6)$
l	=	weyl fermion	$(1, 2, -1/2)$
d^C	=	weyl fermion	$(\bar{3}, 1, 1/3)$
u^C	=	weyl fermion	$(\bar{3}, 1, -2/3)$
e^C	=	weyl fermion	$(1, 1, 1)$
H	=	scalar	$(1, 2, 1/2)$

1.2 Anomaly Summary

Anomalies: **free**.

Channel	Value
$SU(3)_c - SU(3)_c - SU(3)_c$	0
$SU(2)_L - SU(2)_L - U(1)_Y$	0
$SU(3)_c - SU(3)_c - U(1)_Y$	0
$U(1)_Y - gravity$	0
$U(1)_Y - U(1)_Y - U(1)_Y$	0

2 Compact Form of the Full Lagrangian

2.1 Yukawa and Fermion-Mass Terms

$$Y_1 H^\dagger d^C q + \text{h.c.}$$

$$Y_2 u^C q H + \text{h.c.}$$

$$Y_3 H^\dagger e^C l + \text{h.c.}$$

2.2 Scalar Potential

$$m_1^2 H H^\dagger$$

$$\lambda_2 H H H^\dagger H^\dagger$$

2.3 Gauge and Kinetic Terms

$$-\frac{1}{4} B_{\mu\nu} B^{\mu\nu} + -\frac{1}{4} W_{\mu\nu}^a W^{a,\mu\nu} + -\frac{1}{4} G_{\mu\nu}^A G^{A,\mu\nu}$$

$$(D_\mu q)^\dagger D^\mu q + (D_\mu l)^\dagger D^\mu l + (D_\mu d^C)^\dagger D^\mu d^C$$

$$(D_\mu u^C)^\dagger D^\mu u^C + (D_\mu e^C)^\dagger D^\mu e^C + (D_\mu H)^\dagger D^\mu H$$

3 Fully Expanded Form of the Lagrangian

3.1 Yukawa and Fermion-Mass Terms

$$Y_1 (H^{+\dagger} d^{C+1/3, \bar{b}} q_b^{+2/3} + H^{+\dagger} d^{C+1/3, \bar{g}} q_g^{+2/3} + H^{+\dagger} d^{C+1/3, \bar{r}} q_r^{+2/3} + H^{0\dagger} d^{C+1/3, \bar{b}} q_b^{-1/3} + H^{0\dagger} d^{C+1/3, \bar{g}} q_g^{-1/3} + H^{0\dagger} d^{C+1/3, \bar{r}} q_r^{-1/3}) + \text{h.c.}$$

$$Y_2 (u^{C-2/3, \bar{b}} q_b^{+2/3} H^0 - u^{C-2/3, \bar{b}} q_b^{-1/3} H^+ + u^{C-2/3, \bar{g}} q_g^{+2/3} H^0 - u^{C-2/3, \bar{g}} q_g^{-1/3} H^+ + u^{C-2/3, \bar{r}} q_r^{+2/3} H^0 - u^{C-2/3, \bar{r}} q_r^{-1/3} H^+) + \text{h.c.}$$

$$Y_3 (H^{+\dagger} e^{C+} l^0 + H^{0\dagger} e^{C+} l^-) + \text{h.c.}$$

3.2 Scalar Potential

$$m_1^2 (H^0 H^{0\dagger} + H^+ H^{+\dagger})$$

$$\lambda_2 (H^0 H^0 H^{0\dagger} H^{0\dagger} + 2 H^0 H^+ H^{+\dagger} H^{0\dagger} + H^+ H^+ H^{+\dagger} H^{+\dagger})$$

3.3 Gauge and Kinetic Terms

$$\begin{aligned}
& -\frac{1}{4}B_{\mu\nu}B^{\mu\nu} + -\frac{1}{4}W_{\mu\nu}^a W^{a,\mu\nu} + -\frac{1}{4}G_{\mu\nu}^A G^{A,\mu\nu} \\
& \frac{1}{6}g_1 B_\mu q^{+2/3,\bar{r}\dagger} \bar{\sigma}^\mu q_r^{+2/3} + \frac{1}{6}g_1 B_\mu q^{+2/3,\bar{g}\dagger} \bar{\sigma}^\mu q_g^{+2/3} + \frac{1}{6}g_1 B_\mu q^{+2/3,\bar{b}\dagger} \bar{\sigma}^\mu q_b^{+2/3} \\
& \frac{1}{6}g_1 B_\mu q^{-1/3,\bar{r}\dagger} \bar{\sigma}^\mu q_r^{-1/3} + \frac{1}{6}g_1 B_\mu q^{-1/3,\bar{g}\dagger} \bar{\sigma}^\mu q_g^{-1/3} + \frac{1}{6}g_1 B_\mu q^{-1/3,\bar{b}\dagger} \bar{\sigma}^\mu q_b^{-1/3} \\
& \frac{1}{2}g_2 W_\mu^1 q^{+2/3,\bar{r}\dagger} \bar{\sigma}^\mu q_r^{-1/3} + \frac{1}{2}g_2 W_\mu^1 q^{+2/3,\bar{g}\dagger} \bar{\sigma}^\mu q_g^{-1/3} + \frac{1}{2}g_2 W_\mu^1 q^{+2/3,\bar{b}\dagger} \bar{\sigma}^\mu q_b^{-1/3} \\
& \frac{1}{2}g_2 W_\mu^1 q^{-1/3,\bar{r}\dagger} \bar{\sigma}^\mu q_r^{+2/3} + \frac{1}{2}g_2 W_\mu^1 q^{-1/3,\bar{g}\dagger} \bar{\sigma}^\mu q_g^{+2/3} + \frac{1}{2}g_2 W_\mu^1 q^{-1/3,\bar{b}\dagger} \bar{\sigma}^\mu q_b^{+2/3} \\
& \left(-\frac{i}{2}\right) g_2 W_\mu^2 q^{+2/3,\bar{r}\dagger} \bar{\sigma}^\mu q_r^{-1/3} + \left(-\frac{i}{2}\right) g_2 W_\mu^2 q^{+2/3,\bar{g}\dagger} \bar{\sigma}^\mu q_g^{-1/3} + \left(-\frac{i}{2}\right) g_2 W_\mu^2 q^{+2/3,\bar{b}\dagger} \bar{\sigma}^\mu q_b^{-1/3} \\
& \frac{i}{2} g_2 W_\mu^2 q^{-1/3,\bar{r}\dagger} \bar{\sigma}^\mu q_r^{+2/3} + \frac{i}{2} g_2 W_\mu^2 q^{-1/3,\bar{g}\dagger} \bar{\sigma}^\mu q_g^{+2/3} + \frac{i}{2} g_2 W_\mu^2 q^{-1/3,\bar{b}\dagger} \bar{\sigma}^\mu q_b^{+2/3} \\
& \frac{1}{2}g_2 W_\mu^3 q^{+2/3,\bar{r}\dagger} \bar{\sigma}^\mu q_r^{+2/3} + \frac{1}{2}g_2 W_\mu^3 q^{+2/3,\bar{g}\dagger} \bar{\sigma}^\mu q_g^{+2/3} + \frac{1}{2}g_2 W_\mu^3 q^{+2/3,\bar{b}\dagger} \bar{\sigma}^\mu q_b^{+2/3} \\
& \left(-\frac{1}{2}\right) g_2 W_\mu^3 q^{-1/3,\bar{r}\dagger} \bar{\sigma}^\mu q_r^{-1/3} + \left(-\frac{1}{2}\right) g_2 W_\mu^3 q^{-1/3,\bar{g}\dagger} \bar{\sigma}^\mu q_g^{-1/3} + \left(-\frac{1}{2}\right) g_2 W_\mu^3 q^{-1/3,\bar{b}\dagger} \bar{\sigma}^\mu q_b^{-1/3} \\
& \frac{1}{2}g_3 G_\mu^1 q^{+2/3,\bar{r}\dagger} \bar{\sigma}^\mu q_g^{+2/3} + \frac{1}{2}g_3 G_\mu^1 q^{-1/3,\bar{r}\dagger} \bar{\sigma}^\mu q_g^{-1/3} + \frac{1}{2}g_3 G_\mu^1 q^{+2/3,\bar{g}\dagger} \bar{\sigma}^\mu q_r^{+2/3} \\
& \frac{1}{2}g_3 G_\mu^1 q^{-1/3,\bar{g}\dagger} \bar{\sigma}^\mu q_r^{-1/3} + \left(-\frac{i}{2}\right) g_3 G_\mu^2 q^{+2/3,\bar{r}\dagger} \bar{\sigma}^\mu q_g^{+2/3} + \left(-\frac{i}{2}\right) g_3 G_\mu^2 q^{-1/3,\bar{r}\dagger} \bar{\sigma}^\mu q_g^{-1/3} \\
& \frac{i}{2}g_3 G_\mu^2 q^{+2/3,\bar{g}\dagger} \bar{\sigma}^\mu q_r^{+2/3} + \frac{i}{2}g_3 G_\mu^2 q^{-1/3,\bar{g}\dagger} \bar{\sigma}^\mu q_r^{-1/3} + \frac{1}{2}g_3 G_\mu^3 q^{+2/3,\bar{r}\dagger} \bar{\sigma}^\mu q_r^{+2/3} \\
& \frac{1}{2}g_3 G_\mu^3 q^{-1/3,\bar{r}\dagger} \bar{\sigma}^\mu q_r^{-1/3} + \left(-\frac{1}{2}\right) g_3 G_\mu^3 q^{+2/3,\bar{g}\dagger} \bar{\sigma}^\mu q_g^{+2/3} + \left(-\frac{1}{2}\right) g_3 G_\mu^3 q^{-1/3,\bar{g}\dagger} \bar{\sigma}^\mu q_g^{-1/3} \\
& \frac{1}{2}g_3 G_\mu^4 q^{+2/3,\bar{r}\dagger} \bar{\sigma}^\mu q_b^{+2/3} + \frac{1}{2}g_3 G_\mu^4 q^{-1/3,\bar{r}\dagger} \bar{\sigma}^\mu q_b^{-1/3} + \frac{1}{2}g_3 G_\mu^4 q^{+2/3,\bar{b}\dagger} \bar{\sigma}^\mu q_r^{+2/3} \\
& \frac{1}{2}g_3 G_\mu^4 q^{-1/3,\bar{b}\dagger} \bar{\sigma}^\mu q_r^{-1/3} + \left(-\frac{i}{2}\right) g_3 G_\mu^5 q^{+2/3,\bar{r}\dagger} \bar{\sigma}^\mu q_b^{+2/3} + \left(-\frac{i}{2}\right) g_3 G_\mu^5 q^{-1/3,\bar{r}\dagger} \bar{\sigma}^\mu q_b^{-1/3} \\
& \frac{i}{2}g_3 G_\mu^5 q^{+2/3,\bar{b}\dagger} \bar{\sigma}^\mu q_r^{+2/3} + \frac{i}{2}g_3 G_\mu^5 q^{-1/3,\bar{b}\dagger} \bar{\sigma}^\mu q_r^{-1/3} + \frac{1}{2}g_3 G_\mu^6 q^{+2/3,\bar{g}\dagger} \bar{\sigma}^\mu q_b^{+2/3} \\
& \frac{1}{2}g_3 G_\mu^6 q^{-1/3,\bar{g}\dagger} \bar{\sigma}^\mu q_b^{-1/3} + \frac{1}{2}g_3 G_\mu^6 q^{+2/3,\bar{b}\dagger} \bar{\sigma}^\mu q_g^{+2/3} + \frac{1}{2}g_3 G_\mu^6 q^{-1/3,\bar{b}\dagger} \bar{\sigma}^\mu q_g^{-1/3}
\end{aligned}$$

$$\begin{aligned}
& \left(-\frac{i}{2}\right) g_3 G_\mu^7 q^{+2/3, \bar{g}^\dagger} \bar{\sigma}^\mu q_b^{+2/3} + \left(-\frac{i}{2}\right) g_3 G_\mu^7 q^{-1/3, \bar{g}^\dagger} \bar{\sigma}^\mu q_b^{-1/3} + \frac{i}{2} g_3 G_\mu^7 q^{+2/3, \bar{b}^\dagger} \bar{\sigma}^\mu q_g^{+2/3} \\
& \frac{i}{2} g_3 G_\mu^7 q^{-1/3, \bar{b}^\dagger} \bar{\sigma}^\mu q_g^{-1/3} + \frac{\sqrt{3}}{6} g_3 G_\mu^8 q^{+2/3, \bar{r}^\dagger} \bar{\sigma}^\mu q_r^{+2/3} + \frac{\sqrt{3}}{6} g_3 G_\mu^8 q^{-1/3, \bar{r}^\dagger} \bar{\sigma}^\mu q_r^{-1/3} \\
& \frac{\sqrt{3}}{6} g_3 G_\mu^8 q^{+2/3, \bar{g}^\dagger} \bar{\sigma}^\mu q_g^{+2/3} + \frac{\sqrt{3}}{6} g_3 G_\mu^8 q^{-1/3, \bar{g}^\dagger} \bar{\sigma}^\mu q_g^{-1/3} + \left(-\frac{\sqrt{3}}{3}\right) g_3 G_\mu^8 q^{+2/3, \bar{b}^\dagger} \bar{\sigma}^\mu q_b^{+2/3} \\
& \left(-\frac{\sqrt{3}}{3}\right) g_3 G_\mu^8 q^{-1/3, \bar{b}^\dagger} \bar{\sigma}^\mu q_b^{-1/3} + \left(-\frac{1}{2}\right) g_1 B_\mu l^{0\dagger} \bar{\sigma}^\mu l^0 + \left(-\frac{1}{2}\right) g_1 B_\mu l^{-\dagger} \bar{\sigma}^\mu l^- \\
& \frac{1}{2} g_2 W_\mu^1 l^{0\dagger} \bar{\sigma}^\mu l^- + \frac{1}{2} g_2 W_\mu^1 l^{-\dagger} \bar{\sigma}^\mu l^0 + \left(-\frac{i}{2}\right) g_2 W_\mu^2 l^{0\dagger} \bar{\sigma}^\mu l^- \\
& \frac{i}{2} g_2 W_\mu^2 l^{-\dagger} \bar{\sigma}^\mu l^0 + \frac{1}{2} g_2 W_\mu^3 l^{0\dagger} \bar{\sigma}^\mu l^0 + \left(-\frac{1}{2}\right) g_2 W_\mu^3 l^{-\dagger} \bar{\sigma}^\mu l^- \\
& \frac{1}{3} g_1 B_\mu d_r^{C+1/3\dagger} \bar{\sigma}^\mu d^{C+1/3, \bar{r}} + \frac{1}{3} g_1 B_\mu d_g^{C+1/3\dagger} \bar{\sigma}^\mu d^{C+1/3, \bar{g}} + \frac{1}{3} g_1 B_\mu d_b^{C+1/3\dagger} \bar{\sigma}^\mu d^{C+1/3, \bar{b}} \\
& \left(-\frac{1}{2}\right) g_3 G_\mu^1 d_r^{C+1/3\dagger} \bar{\sigma}^\mu d^{C+1/3, \bar{g}} + \left(-\frac{1}{2}\right) g_3 G_\mu^1 d_g^{C+1/3\dagger} \bar{\sigma}^\mu d^{C+1/3, \bar{r}} + \left(-\frac{i}{2}\right) g_3 G_\mu^2 d_r^{C+1/3\dagger} \bar{\sigma}^\mu d^{C+1/3, \bar{b}} \\
& \frac{i}{2} g_3 G_\mu^2 d_g^{C+1/3\dagger} \bar{\sigma}^\mu d^{C+1/3, \bar{r}} + \left(-\frac{1}{2}\right) g_3 G_\mu^3 d_r^{C+1/3\dagger} \bar{\sigma}^\mu d^{C+1/3, \bar{b}} + \frac{1}{2} g_3 G_\mu^3 d_g^{C+1/3\dagger} \bar{\sigma}^\mu d^{C+1/3, \bar{g}} \\
& \left(-\frac{1}{2}\right) g_3 G_\mu^4 d_r^{C+1/3\dagger} \bar{\sigma}^\mu d^{C+1/3, \bar{b}} + \left(-\frac{1}{2}\right) g_3 G_\mu^4 d_b^{C+1/3\dagger} \bar{\sigma}^\mu d^{C+1/3, \bar{r}} + \left(-\frac{i}{2}\right) g_3 G_\mu^5 d_r^{C+1/3\dagger} \bar{\sigma}^\mu d^{C+1/3, \bar{g}} \\
& \frac{i}{2} g_3 G_\mu^5 d_b^{C+1/3\dagger} \bar{\sigma}^\mu d^{C+1/3, \bar{r}} + \left(-\frac{1}{2}\right) g_3 G_\mu^6 d_g^{C+1/3\dagger} \bar{\sigma}^\mu d^{C+1/3, \bar{b}} + \left(-\frac{1}{2}\right) g_3 G_\mu^6 d_b^{C+1/3\dagger} \bar{\sigma}^\mu d^{C+1/3, \bar{g}} \\
& \left(-\frac{i}{2}\right) g_3 G_\mu^7 d_g^{C+1/3\dagger} \bar{\sigma}^\mu d^{C+1/3, \bar{b}} + \frac{i}{2} g_3 G_\mu^7 d_b^{C+1/3\dagger} \bar{\sigma}^\mu d^{C+1/3, \bar{g}} + \left(-\frac{\sqrt{3}}{6}\right) g_3 G_\mu^8 d_r^{C+1/3\dagger} \bar{\sigma}^\mu d^{C+1/3, \bar{r}} \\
& \left(-\frac{\sqrt{3}}{6}\right) g_3 G_\mu^8 d_g^{C+1/3\dagger} \bar{\sigma}^\mu d^{C+1/3, \bar{g}} + \frac{\sqrt{3}}{3} g_3 G_\mu^8 d_b^{C+1/3\dagger} \bar{\sigma}^\mu d^{C+1/3, \bar{b}} + \left(-\frac{2}{3}\right) g_1 B_\mu u_r^{C-2/3\dagger} \bar{\sigma}^\mu u^{C-2/3, \bar{r}} \\
& \left(-\frac{2}{3}\right) g_1 B_\mu u_g^{C-2/3\dagger} \bar{\sigma}^\mu u^{C-2/3, \bar{g}} + \left(-\frac{2}{3}\right) g_1 B_\mu u_b^{C-2/3\dagger} \bar{\sigma}^\mu u^{C-2/3, \bar{b}} + \left(-\frac{1}{2}\right) g_3 G_\mu^1 u_r^{C-2/3\dagger} \bar{\sigma}^\mu u^{C-2/3, \bar{g}} \\
& \left(-\frac{1}{2}\right) g_3 G_\mu^1 u_g^{C-2/3\dagger} \bar{\sigma}^\mu u^{C-2/3, \bar{r}} + \left(-\frac{i}{2}\right) g_3 G_\mu^2 u_r^{C-2/3\dagger} \bar{\sigma}^\mu u^{C-2/3, \bar{b}} + \frac{i}{2} g_3 G_\mu^2 u_g^{C-2/3\dagger} \bar{\sigma}^\mu u^{C-2/3, \bar{g}} \\
& \left(-\frac{1}{2}\right) g_3 G_\mu^3 u_r^{C-2/3\dagger} \bar{\sigma}^\mu u^{C-2/3, \bar{r}} + \frac{1}{2} g_3 G_\mu^3 u_g^{C-2/3\dagger} \bar{\sigma}^\mu u^{C-2/3, \bar{b}} + \left(-\frac{1}{2}\right) g_3 G_\mu^4 u_r^{C-2/3\dagger} \bar{\sigma}^\mu u^{C-2/3, \bar{g}}
\end{aligned}$$

$$\left(-\frac{1}{2}\right) g_3 G_\mu^4 u_b^{C-2/3\dagger} \bar{\sigma}^\mu u^{C-2/3,\bar{r}} + \left(-\frac{i}{2}\right) g_3 G_\mu^5 u_r^{C-2/3\dagger} \bar{\sigma}^\mu u^{C-2/3,\bar{b}} + \frac{i}{2} g_3 G_\mu^5 u_b^{C-2/3\dagger} \bar{\sigma}^\mu u^{C-2/3,\bar{r}}$$

$$\left(-\frac{1}{2}\right) g_3 G_\mu^6 u_g^{C-2/3\dagger} \bar{\sigma}^\mu u^{C-2/3,\bar{b}} + \left(-\frac{1}{2}\right) g_3 G_\mu^6 u_b^{C-2/3\dagger} \bar{\sigma}^\mu u^{C-2/3,\bar{g}} + \left(-\frac{i}{2}\right) g_3 G_\mu^7 u_g^{C-2/3\dagger} \bar{\sigma}^\mu u^{C-2/3,\bar{b}}$$

$$\frac{i}{2} g_3 G_\mu^7 u_b^{C-2/3\dagger} \bar{\sigma}^\mu u^{C-2/3,\bar{g}} + \left(-\frac{\sqrt{3}}{6}\right) g_3 G_\mu^8 u_r^{C-2/3\dagger} \bar{\sigma}^\mu u^{C-2/3,\bar{r}} + \left(-\frac{\sqrt{3}}{6}\right) g_3 G_\mu^8 u_g^{C-2/3\dagger} \bar{\sigma}^\mu u^{C-2/3,\bar{g}}$$

$$\frac{\sqrt{3}}{3} g_3 G_\mu^8 u_b^{C-2/3\dagger} \bar{\sigma}^\mu u^{C-2/3,\bar{b}} + g_1 B_\mu e^{C+\dagger} \bar{\sigma}^\mu e^{C+} + \frac{1}{2} g_1 B_\mu H^{+\dagger} i \overleftrightarrow{\partial}^\mu H^+$$

$$\frac{1}{2} g_1 B_\mu H^{0\dagger} i \overleftrightarrow{\partial}^\mu H^0 + \frac{1}{2} g_2 W_\mu^1 H^{+\dagger} i \overleftrightarrow{\partial}^\mu H^0 + \frac{1}{2} g_2 W_\mu^1 H^{0\dagger} i \overleftrightarrow{\partial}^\mu H^+$$

$$\left(-\frac{i}{2}\right) g_2 W_\mu^2 H^{+\dagger} i \overleftrightarrow{\partial}^\mu H^0 + \frac{i}{2} g_2 W_\mu^2 H^{0\dagger} i \overleftrightarrow{\partial}^\mu H^+ + \frac{1}{2} g_2 W_\mu^3 H^{+\dagger} i \overleftrightarrow{\partial}^\mu H^+$$

$$\left(-\frac{1}{2}\right) g_2 W_\mu^3 H^{0\dagger} i \overleftrightarrow{\partial}^\mu H^0$$

$$\frac{1}{4} g_1 g_1 B_\mu B_\mu H^{+\dagger} H^+ + \frac{1}{4} g_1 g_1 B_\mu B_\mu H^{0\dagger} H^0 + \frac{1}{4} g_1 g_2 B_\mu W_\mu^1 H^{+\dagger} H^0$$

$$\frac{1}{4} g_1 g_2 B_\mu W_\mu^1 H^{0\dagger} H^+ + \left(-\frac{i}{4}\right) g_1 g_2 B_\mu W_\mu^2 H^{+\dagger} H^0 + \frac{i}{4} g_1 g_2 B_\mu W_\mu^2 H^{0\dagger} H^+$$

$$\frac{1}{4} g_1 g_2 B_\mu W_\mu^3 H^{+\dagger} H^+ + \left(-\frac{1}{4}\right) g_1 g_2 B_\mu W_\mu^3 H^{0\dagger} H^0 + \frac{1}{4} g_2 g_1 W_\mu^1 B_\mu H^{+\dagger} H^0$$

$$\frac{1}{4} g_2 g_1 W_\mu^1 B_\mu H^{0\dagger} H^+ + \left(-\frac{i}{4}\right) g_2 g_1 W_\mu^2 B_\mu H^{+\dagger} H^0 + \frac{i}{4} g_2 g_1 W_\mu^2 B_\mu H^{0\dagger} H^+$$

$$\frac{1}{4} g_2 g_1 W_\mu^3 B_\mu H^{+\dagger} H^+ + \left(-\frac{1}{4}\right) g_2 g_1 W_\mu^3 B_\mu H^{0\dagger} H^0 + \frac{1}{4} g_2 g_2 W_\mu^1 W_\mu^1 H^{+\dagger} H^+$$

$$\frac{1}{4} g_2 g_2 W_\mu^1 W_\mu^1 H^{0\dagger} H^0 + \frac{i}{4} g_2 g_2 W_\mu^1 W_\mu^2 H^{+\dagger} H^+ + \left(-\frac{i}{4}\right) g_2 g_2 W_\mu^1 W_\mu^2 H^{0\dagger} H^0$$

$$\left(-\frac{1}{4}\right) g_2 g_2 W_\mu^1 W_\mu^3 H^{+\dagger} H^0 + \frac{1}{4} g_2 g_2 W_\mu^1 W_\mu^3 H^{0\dagger} H^+ + \left(-\frac{i}{4}\right) g_2 g_2 W_\mu^2 W_\mu^1 H^{+\dagger} H^+$$

$$\frac{i}{4} g_2 g_2 W_\mu^2 W_\mu^1 H^{0\dagger} H^0 + \frac{1}{4} g_2 g_2 W_\mu^2 W_\mu^2 H^{+\dagger} H^+ + \frac{1}{4} g_2 g_2 W_\mu^2 W_\mu^2 H^{0\dagger} H^0$$

$$\frac{i}{4} g_2 g_2 W_\mu^2 W_\mu^3 H^{+\dagger} H^0 + \frac{i}{4} g_2 g_2 W_\mu^2 W_\mu^3 H^{0\dagger} H^+ + \frac{1}{4} g_2 g_2 W_\mu^3 W_\mu^1 H^{+\dagger} H^0$$

$$\begin{aligned}
& \left(-\frac{1}{4}\right) g_2 g_2 W_\mu^3 W_\mu^1 H^{0\dagger} H^+ + \left(-\frac{i}{4}\right) g_2 g_2 W_\mu^3 W_\mu^2 H^{+\dagger} H^0 + \left(-\frac{i}{4}\right) g_2 g_2 W_\mu^3 W_\mu^2 H^{0\dagger} H^+ \\
& \frac{1}{4} g_2 g_2 W_\mu^3 W_\mu^3 H^{+\dagger} H^+ + \frac{1}{4} g_2 g_2 W_\mu^3 W_\mu^3 H^{0\dagger} H^0 \\
& -g_2 (\partial_\mu W_\nu^1) W^{2\mu} W^{3\nu} + -g_2 (\partial_\mu W_\nu^2) W^{3\mu} W^{1\nu} + -g_2 (\partial_\mu W_\nu^3) W^{1\mu} W^{2\nu} \\
& g_2 (\partial_\mu W_\nu^2) W^{1\mu} W^{3\nu} + g_2 (\partial_\mu W_\nu^3) W^{2\mu} W^{1\nu} + g_2 (\partial_\mu W_\nu^1) W^{3\mu} W^{2\nu} \\
& -\frac{g_2^2}{4} W_\mu^1 W_\nu^2 W^{1\mu} W^{2\nu} + \frac{g_2^2}{4} W_\mu^1 W_\nu^2 W^{2\mu} W^{1\nu} + -\frac{g_2^2}{4} W_\mu^2 W_\nu^3 W^{2\mu} W^{3\nu} \\
& \frac{g_2^2}{4} W_\mu^2 W_\nu^3 W^{3\mu} W^{2\nu} + -\frac{g_2^2}{4} W_\mu^3 W_\nu^1 W^{3\mu} W^{1\nu} + \frac{g_2^2}{4} W_\mu^3 W_\nu^1 W^{1\mu} W^{3\nu} \\
& \frac{g_2^2}{4} W_\mu^2 W_\nu^1 W^{1\mu} W^{2\nu} + -\frac{g_2^2}{4} W_\mu^2 W_\nu^1 W^{2\mu} W^{1\nu} + \frac{g_2^2}{4} W_\mu^3 W_\nu^2 W^{2\mu} W^{3\nu} \\
& -\frac{g_2^2}{4} W_\mu^3 W_\nu^2 W^{3\mu} W^{2\nu} + \frac{g_2^2}{4} W_\mu^1 W_\nu^3 W^{3\mu} W^{1\nu} + -\frac{g_2^2}{4} W_\mu^1 W_\nu^3 W^{1\mu} W^{3\nu} \\
& -g_3 (\partial_\mu G_\nu^1) G^{2\mu} G^{3\nu} + g_3 (\partial_\mu G_\nu^1) G^{3\mu} G^{2\nu} + -\frac{g_3}{2} (\partial_\mu G_\nu^1) G^{4\mu} G^{7\nu} \\
& \frac{g_3}{2} (\partial_\mu G_\nu^1) G^{5\mu} G^{6\nu} + -\frac{g_3}{2} (\partial_\mu G_\nu^1) G^{6\mu} G^{5\nu} + \frac{g_3}{2} (\partial_\mu G_\nu^1) G^{7\mu} G^{4\nu} \\
& g_3 (\partial_\mu G_\nu^2) G^{1\mu} G^{3\nu} + -g_3 (\partial_\mu G_\nu^2) G^{3\mu} G^{1\nu} + -\frac{g_3}{2} (\partial_\mu G_\nu^2) G^{4\mu} G^{6\nu} \\
& -\frac{g_3}{2} (\partial_\mu G_\nu^2) G^{5\mu} G^{7\nu} + \frac{g_3}{2} (\partial_\mu G_\nu^2) G^{6\mu} G^{4\nu} + \frac{g_3}{2} (\partial_\mu G_\nu^2) G^{7\mu} G^{5\nu} \\
& -g_3 (\partial_\mu G_\nu^3) G^{1\mu} G^{2\nu} + g_3 (\partial_\mu G_\nu^3) G^{2\mu} G^{1\nu} + -\frac{g_3}{2} (\partial_\mu G_\nu^3) G^{4\mu} G^{5\nu} \\
& \frac{g_3}{2} (\partial_\mu G_\nu^3) G^{5\mu} G^{4\nu} + \frac{g_3}{2} (\partial_\mu G_\nu^3) G^{6\mu} G^{7\nu} + -\frac{g_3}{2} (\partial_\mu G_\nu^3) G^{7\mu} G^{6\nu} \\
& \frac{g_3}{2} (\partial_\mu G_\nu^4) G^{1\mu} G^{7\nu} + \frac{g_3}{2} (\partial_\mu G_\nu^4) G^{2\mu} G^{6\nu} + \frac{g_3}{2} (\partial_\mu G_\nu^4) G^{3\mu} G^{5\nu} \\
& -\frac{g_3}{2} (\partial_\mu G_\nu^4) G^{5\mu} G^{3\nu} + -\frac{\sqrt{3}g_3}{2} (\partial_\mu G_\nu^4) G^{5\mu} G^{8\nu} + -\frac{g_3}{2} (\partial_\mu G_\nu^4) G^{6\mu} G^{2\nu} \\
& -\frac{g_3}{2} (\partial_\mu G_\nu^4) G^{7\mu} G^{1\nu} + \frac{\sqrt{3}g_3}{2} (\partial_\mu G_\nu^4) G^{8\mu} G^{5\nu} + -\frac{g_3}{2} (\partial_\mu G_\nu^5) G^{1\mu} G^{6\nu} \\
& \frac{g_3}{2} (\partial_\mu G_\nu^5) G^{2\mu} G^{7\nu} + -\frac{g_3}{2} (\partial_\mu G_\nu^5) G^{3\mu} G^{4\nu} + \frac{g_3}{2} (\partial_\mu G_\nu^5) G^{4\mu} G^{3\nu}
\end{aligned}$$

$$\begin{aligned}
& \frac{\sqrt{3}g_3}{2} (\partial_\mu G_\nu^5) G^{4\mu} G^{8\nu} + \frac{g_3}{2} (\partial_\mu G_\nu^5) G^{6\mu} G^{1\nu} + -\frac{g_3}{2} (\partial_\mu G_\nu^5) G^{7\mu} G^{2\nu} \\
& -\frac{\sqrt{3}g_3}{2} (\partial_\mu G_\nu^5) G^{8\mu} G^{4\nu} + \frac{g_3}{2} (\partial_\mu G_\nu^6) G^{1\mu} G^{5\nu} + -\frac{g_3}{2} (\partial_\mu G_\nu^6) G^{2\mu} G^{4\nu} \\
& -\frac{g_3}{2} (\partial_\mu G_\nu^6) G^{3\mu} G^{7\nu} + \frac{g_3}{2} (\partial_\mu G_\nu^6) G^{4\mu} G^{2\nu} + -\frac{g_3}{2} (\partial_\mu G_\nu^6) G^{5\mu} G^{1\nu} \\
& \frac{g_3}{2} (\partial_\mu G_\nu^6) G^{7\mu} G^{3\nu} + -\frac{\sqrt{3}g_3}{2} (\partial_\mu G_\nu^6) G^{7\mu} G^{8\nu} + \frac{\sqrt{3}g_3}{2} (\partial_\mu G_\nu^6) G^{8\mu} G^{7\nu} \\
& -\frac{g_3}{2} (\partial_\mu G_\nu^7) G^{1\mu} G^{4\nu} + -\frac{g_3}{2} (\partial_\mu G_\nu^7) G^{2\mu} G^{5\nu} + \frac{g_3}{2} (\partial_\mu G_\nu^7) G^{3\mu} G^{6\nu} \\
& \frac{g_3}{2} (\partial_\mu G_\nu^7) G^{4\mu} G^{1\nu} + \frac{g_3}{2} (\partial_\mu G_\nu^7) G^{5\mu} G^{2\nu} + -\frac{g_3}{2} (\partial_\mu G_\nu^7) G^{6\mu} G^{3\nu} \\
& \frac{\sqrt{3}g_3}{2} (\partial_\mu G_\nu^7) G^{6\mu} G^{8\nu} + -\frac{\sqrt{3}g_3}{2} (\partial_\mu G_\nu^7) G^{8\mu} G^{6\nu} + -\frac{\sqrt{3}g_3}{2} (\partial_\mu G_\nu^8) G^{4\mu} G^{5\nu} \\
& \frac{\sqrt{3}g_3}{2} (\partial_\mu G_\nu^8) G^{5\mu} G^{4\nu} + -\frac{\sqrt{3}g_3}{2} (\partial_\mu G_\nu^8) G^{6\mu} G^{7\nu} + \frac{\sqrt{3}g_3}{2} (\partial_\mu G_\nu^8) G^{7\mu} G^{6\nu} \\
& -\frac{g_3^2}{4} G_\mu^1 G_\nu^2 G^{1\mu} G^{2\nu} + \frac{g_3^2}{4} G_\mu^1 G_\nu^2 G^{2\mu} G^{1\nu} + -\frac{g_3^2}{8} G_\mu^1 G_\nu^2 G^{4\mu} G^{5\nu} \\
& \frac{g_3^2}{8} G_\mu^1 G_\nu^2 G^{5\mu} G^{4\nu} + \frac{g_3^2}{8} G_\mu^1 G_\nu^2 G^{6\mu} G^{7\nu} + -\frac{g_3^2}{8} G_\mu^1 G_\nu^2 G^{7\mu} G^{6\nu} \\
& -\frac{g_3^2}{4} G_\mu^1 G_\nu^3 G^{1\mu} G^{3\nu} + \frac{g_3^2}{4} G_\mu^1 G_\nu^3 G^{3\mu} G^{1\nu} + \frac{g_3^2}{8} G_\mu^1 G_\nu^3 G^{4\mu} G^{6\nu} \\
& \frac{g_3^2}{8} G_\mu^1 G_\nu^3 G^{5\mu} G^{7\nu} + -\frac{g_3^2}{8} G_\mu^1 G_\nu^3 G^{6\mu} G^{4\nu} + -\frac{g_3^2}{8} G_\mu^1 G_\nu^3 G^{7\mu} G^{5\nu} \\
& -\frac{g_3^2}{16} G_\mu^1 G_\nu^4 G^{1\mu} G^{4\nu} + -\frac{g_3^2}{16} G_\mu^1 G_\nu^4 G^{2\mu} G^{5\nu} + \frac{g_3^2}{16} G_\mu^1 G_\nu^4 G^{3\mu} G^{6\nu} \\
& \frac{g_3^2}{16} G_\mu^1 G_\nu^4 G^{4\mu} G^{1\nu} + \frac{g_3^2}{16} G_\mu^1 G_\nu^4 G^{5\mu} G^{2\nu} + -\frac{g_3^2}{16} G_\mu^1 G_\nu^4 G^{6\mu} G^{3\nu} \\
& \frac{\sqrt{3}g_3^2}{16} G_\mu^1 G_\nu^4 G^{6\mu} G^{8\nu} + -\frac{\sqrt{3}g_3^2}{16} G_\mu^1 G_\nu^4 G^{8\mu} G^{6\nu} + -\frac{g_3^2}{16} G_\mu^1 G_\nu^5 G^{1\mu} G^{5\nu} \\
& \frac{g_3^2}{16} G_\mu^1 G_\nu^5 G^{2\mu} G^{4\nu} + \frac{g_3^2}{16} G_\mu^1 G_\nu^5 G^{3\mu} G^{7\nu} + -\frac{g_3^2}{16} G_\mu^1 G_\nu^5 G^{4\mu} G^{2\nu} \\
& \frac{g_3^2}{16} G_\mu^1 G_\nu^5 G^{5\mu} G^{1\nu} + -\frac{g_3^2}{16} G_\mu^1 G_\nu^5 G^{7\mu} G^{3\nu} + \frac{\sqrt{3}g_3^2}{16} G_\mu^1 G_\nu^5 G^{7\mu} G^{8\nu}
\end{aligned}$$

$$\begin{aligned}
& -\frac{\sqrt{3}g_3^2}{16} G_\mu^1 G_\nu^5 G^{8\mu} G^{7\nu} + -\frac{g_3^2}{16} G_\mu^1 G_\nu^6 G^{1\mu} G^{6\nu} + \frac{g_3^2}{16} G_\mu^1 G_\nu^6 G^{2\mu} G^{7\nu} \\
& -\frac{g_3^2}{16} G_\mu^1 G_\nu^6 G^{3\mu} G^{4\nu} + \frac{g_3^2}{16} G_\mu^1 G_\nu^6 G^{4\mu} G^{3\nu} + \frac{\sqrt{3}g_3^2}{16} G_\mu^1 G_\nu^6 G^{4\mu} G^{8\nu} \\
& \frac{g_3^2}{16} G_\mu^1 G_\nu^6 G^{6\mu} G^{1\nu} + -\frac{g_3^2}{16} G_\mu^1 G_\nu^6 G^{7\mu} G^{2\nu} + -\frac{\sqrt{3}g_3^2}{16} G_\mu^1 G_\nu^6 G^{8\mu} G^{4\nu} \\
& -\frac{g_3^2}{16} G_\mu^1 G_\nu^7 G^{1\mu} G^{7\nu} + -\frac{g_3^2}{16} G_\mu^1 G_\nu^7 G^{2\mu} G^{6\nu} + -\frac{g_3^2}{16} G_\mu^1 G_\nu^7 G^{3\mu} G^{5\nu} \\
& \frac{g_3^2}{16} G_\mu^1 G_\nu^7 G^{5\mu} G^{3\nu} + \frac{\sqrt{3}g_3^2}{16} G_\mu^1 G_\nu^7 G^{5\mu} G^{8\nu} + \frac{g_3^2}{16} G_\mu^1 G_\nu^7 G^{6\mu} G^{2\nu} \\
& \frac{g_3^2}{16} G_\mu^1 G_\nu^7 G^{7\mu} G^{1\nu} + -\frac{\sqrt{3}g_3^2}{16} G_\mu^1 G_\nu^7 G^{8\mu} G^{5\nu} + \frac{g_3^2}{4} G_\mu^2 G_\nu^1 G^{1\mu} G^{2\nu} \\
& -\frac{g_3^2}{4} G_\mu^2 G_\nu^1 G^{2\mu} G^{1\nu} + \frac{g_3^2}{8} G_\mu^2 G_\nu^1 G^{4\mu} G^{5\nu} + -\frac{g_3^2}{8} G_\mu^2 G_\nu^1 G^{5\mu} G^{4\nu} \\
& -\frac{g_3^2}{8} G_\mu^2 G_\nu^1 G^{6\mu} G^{7\nu} + \frac{g_3^2}{8} G_\mu^2 G_\nu^1 G^{7\mu} G^{6\nu} + -\frac{g_3^2}{4} G_\mu^2 G_\nu^3 G^{2\mu} G^{3\nu} \\
& \frac{g_3^2}{4} G_\mu^2 G_\nu^3 G^{3\mu} G^{2\nu} + -\frac{g_3^2}{8} G_\mu^2 G_\nu^3 G^{4\mu} G^{7\nu} + \frac{g_3^2}{8} G_\mu^2 G_\nu^3 G^{5\mu} G^{6\nu} \\
& -\frac{g_3^2}{8} G_\mu^2 G_\nu^3 G^{6\mu} G^{5\nu} + \frac{g_3^2}{8} G_\mu^2 G_\nu^3 G^{7\mu} G^{4\nu} + \frac{g_3^2}{16} G_\mu^2 G_\nu^4 G^{1\mu} G^{5\nu} \\
& -\frac{g_3^2}{16} G_\mu^2 G_\nu^4 G^{2\mu} G^{4\nu} + -\frac{g_3^2}{16} G_\mu^2 G_\nu^4 G^{3\mu} G^{7\nu} + \frac{g_3^2}{16} G_\mu^2 G_\nu^4 G^{4\mu} G^{2\nu} \\
& -\frac{g_3^2}{16} G_\mu^2 G_\nu^4 G^{5\mu} G^{1\nu} + \frac{g_3^2}{16} G_\mu^2 G_\nu^4 G^{7\mu} G^{3\nu} + -\frac{\sqrt{3}g_3^2}{16} G_\mu^2 G_\nu^4 G^{7\mu} G^{8\nu} \\
& \frac{\sqrt{3}g_3^2}{16} G_\mu^2 G_\nu^4 G^{8\mu} G^{7\nu} + -\frac{g_3^2}{16} G_\mu^2 G_\nu^5 G^{1\mu} G^{4\nu} + -\frac{g_3^2}{16} G_\mu^2 G_\nu^5 G^{2\mu} G^{5\nu} \\
& \frac{g_3^2}{16} G_\mu^2 G_\nu^5 G^{3\mu} G^{6\nu} + \frac{g_3^2}{16} G_\mu^2 G_\nu^5 G^{4\mu} G^{1\nu} + \frac{g_3^2}{16} G_\mu^2 G_\nu^5 G^{5\mu} G^{2\nu} \\
& -\frac{g_3^2}{16} G_\mu^2 G_\nu^5 G^{6\mu} G^{3\nu} + \frac{\sqrt{3}g_3^2}{16} G_\mu^2 G_\nu^5 G^{6\mu} G^{8\nu} + -\frac{\sqrt{3}g_3^2}{16} G_\mu^2 G_\nu^5 G^{8\mu} G^{6\nu} \\
& -\frac{g_3^2}{16} G_\mu^2 G_\nu^6 G^{1\mu} G^{7\nu} + -\frac{g_3^2}{16} G_\mu^2 G_\nu^6 G^{2\mu} G^{6\nu} + -\frac{g_3^2}{16} G_\mu^2 G_\nu^6 G^{3\mu} G^{5\nu} \\
& \frac{g_3^2}{16} G_\mu^2 G_\nu^6 G^{5\mu} G^{3\nu} + \frac{\sqrt{3}g_3^2}{16} G_\mu^2 G_\nu^6 G^{5\mu} G^{8\nu} + \frac{g_3^2}{16} G_\mu^2 G_\nu^6 G^{6\mu} G^{2\nu}
\end{aligned}$$

$$\begin{aligned}
& \frac{g_3^2}{16} G_\mu^2 G_\nu^6 G^{7\mu} G^{1\nu} + -\frac{\sqrt{3}g_3^2}{16} G_\mu^2 G_\nu^6 G^{8\mu} G^{5\nu} + \frac{g_3^2}{16} G_\mu^2 G_\nu^7 G^{1\mu} G^{6\nu} \\
& -\frac{g_3^2}{16} G_\mu^2 G_\nu^7 G^{2\mu} G^{7\nu} + \frac{g_3^2}{16} G_\mu^2 G_\nu^7 G^{3\mu} G^{4\nu} + -\frac{g_3^2}{16} G_\mu^2 G_\nu^7 G^{4\mu} G^{3\nu} \\
& -\frac{\sqrt{3}g_3^2}{16} G_\mu^2 G_\nu^7 G^{4\mu} G^{8\nu} + -\frac{g_3^2}{16} G_\mu^2 G_\nu^7 G^{6\mu} G^{1\nu} + \frac{g_3^2}{16} G_\mu^2 G_\nu^7 G^{7\mu} G^{2\nu} \\
& \frac{\sqrt{3}g_3^2}{16} G_\mu^2 G_\nu^7 G^{8\mu} G^{4\nu} + \frac{g_3^2}{4} G_\mu^3 G_\nu^1 G^{1\mu} G^{3\nu} + -\frac{g_3^2}{4} G_\mu^3 G_\nu^1 G^{3\mu} G^{1\nu} \\
& -\frac{g_3^2}{8} G_\mu^3 G_\nu^1 G^{4\mu} G^{6\nu} + -\frac{g_3^2}{8} G_\mu^3 G_\nu^1 G^{5\mu} G^{7\nu} + \frac{g_3^2}{8} G_\mu^3 G_\nu^1 G^{6\mu} G^{4\nu} \\
& \frac{g_3^2}{8} G_\mu^3 G_\nu^1 G^{7\mu} G^{5\nu} + \frac{g_3^2}{4} G_\mu^3 G_\nu^2 G^{2\mu} G^{3\nu} + -\frac{g_3^2}{4} G_\mu^3 G_\nu^2 G^{3\mu} G^{2\nu} \\
& \frac{g_3^2}{8} G_\mu^3 G_\nu^2 G^{4\mu} G^{7\nu} + -\frac{g_3^2}{8} G_\mu^3 G_\nu^2 G^{5\mu} G^{6\nu} + \frac{g_3^2}{8} G_\mu^3 G_\nu^2 G^{6\mu} G^{5\nu} \\
& -\frac{g_3^2}{8} G_\mu^3 G_\nu^2 G^{7\mu} G^{4\nu} + -\frac{g_3^2}{16} G_\mu^3 G_\nu^4 G^{1\mu} G^{6\nu} + \frac{g_3^2}{16} G_\mu^3 G_\nu^4 G^{2\mu} G^{7\nu} \\
& -\frac{g_3^2}{16} G_\mu^3 G_\nu^4 G^{3\mu} G^{4\nu} + \frac{g_3^2}{16} G_\mu^3 G_\nu^4 G^{4\mu} G^{3\nu} + \frac{\sqrt{3}g_3^2}{16} G_\mu^3 G_\nu^4 G^{4\mu} G^{8\nu} \\
& \frac{g_3^2}{16} G_\mu^3 G_\nu^4 G^{6\mu} G^{1\nu} + -\frac{g_3^2}{16} G_\mu^3 G_\nu^4 G^{7\mu} G^{2\nu} + -\frac{\sqrt{3}g_3^2}{16} G_\mu^3 G_\nu^4 G^{8\mu} G^{4\nu} \\
& -\frac{g_3^2}{16} G_\mu^3 G_\nu^5 G^{1\mu} G^{7\nu} + -\frac{g_3^2}{16} G_\mu^3 G_\nu^5 G^{2\mu} G^{6\nu} + -\frac{g_3^2}{16} G_\mu^3 G_\nu^5 G^{3\mu} G^{5\nu} \\
& \frac{g_3^2}{16} G_\mu^3 G_\nu^5 G^{5\mu} G^{3\nu} + \frac{\sqrt{3}g_3^2}{16} G_\mu^3 G_\nu^5 G^{5\mu} G^{8\nu} + \frac{g_3^2}{16} G_\mu^3 G_\nu^5 G^{6\mu} G^{2\nu} \\
& \frac{g_3^2}{16} G_\mu^3 G_\nu^5 G^{7\mu} G^{1\nu} + -\frac{\sqrt{3}g_3^2}{16} G_\mu^3 G_\nu^5 G^{8\mu} G^{5\nu} + \frac{g_3^2}{16} G_\mu^3 G_\nu^6 G^{1\mu} G^{4\nu} \\
& \frac{g_3^2}{16} G_\mu^3 G_\nu^6 G^{2\mu} G^{5\nu} + -\frac{g_3^2}{16} G_\mu^3 G_\nu^6 G^{3\mu} G^{6\nu} + -\frac{g_3^2}{16} G_\mu^3 G_\nu^6 G^{4\mu} G^{1\nu} \\
& -\frac{g_3^2}{16} G_\mu^3 G_\nu^6 G^{5\mu} G^{2\nu} + \frac{g_3^2}{16} G_\mu^3 G_\nu^6 G^{6\mu} G^{3\nu} + -\frac{\sqrt{3}g_3^2}{16} G_\mu^3 G_\nu^6 G^{6\mu} G^{8\nu} \\
& \frac{\sqrt{3}g_3^2}{16} G_\mu^3 G_\nu^6 G^{8\mu} G^{6\nu} + \frac{g_3^2}{16} G_\mu^3 G_\nu^7 G^{1\mu} G^{5\nu} + -\frac{g_3^2}{16} G_\mu^3 G_\nu^7 G^{2\mu} G^{4\nu} \\
& -\frac{g_3^2}{16} G_\mu^3 G_\nu^7 G^{3\mu} G^{7\nu} + \frac{g_3^2}{16} G_\mu^3 G_\nu^7 G^{4\mu} G^{2\nu} + -\frac{g_3^2}{16} G_\mu^3 G_\nu^7 G^{5\mu} G^{1\nu}
\end{aligned}$$

$$\begin{aligned}
& \frac{g_3^2}{16} G_\mu^3 G_\nu^7 G^{7\mu} G^{3\nu} + -\frac{\sqrt{3}g_3^2}{16} G_\mu^3 G_\nu^7 G^{7\mu} G^{8\nu} + \frac{\sqrt{3}g_3^2}{16} G_\mu^3 G_\nu^7 G^{8\mu} G^{7\nu} \\
& \frac{g_3^2}{16} G_\mu^4 G_\nu^1 G^{1\mu} G^{4\nu} + \frac{g_3^2}{16} G_\mu^4 G_\nu^1 G^{2\mu} G^{5\nu} + -\frac{g_3^2}{16} G_\mu^4 G_\nu^1 G^{3\mu} G^{6\nu} \\
& -\frac{g_3^2}{16} G_\mu^4 G_\nu^1 G^{4\mu} G^{1\nu} + -\frac{g_3^2}{16} G_\mu^4 G_\nu^1 G^{5\mu} G^{2\nu} + \frac{g_3^2}{16} G_\mu^4 G_\nu^1 G^{6\mu} G^{3\nu} \\
& -\frac{\sqrt{3}g_3^2}{16} G_\mu^4 G_\nu^1 G^{6\mu} G^{8\nu} + \frac{\sqrt{3}g_3^2}{16} G_\mu^4 G_\nu^1 G^{8\mu} G^{6\nu} + -\frac{g_3^2}{16} G_\mu^4 G_\nu^2 G^{1\mu} G^{5\nu} \\
& \frac{g_3^2}{16} G_\mu^4 G_\nu^2 G^{2\mu} G^{4\nu} + \frac{g_3^2}{16} G_\mu^4 G_\nu^2 G^{3\mu} G^{7\nu} + -\frac{g_3^2}{16} G_\mu^4 G_\nu^2 G^{4\mu} G^{2\nu} \\
& \frac{g_3^2}{16} G_\mu^4 G_\nu^2 G^{5\mu} G^{1\nu} + -\frac{g_3^2}{16} G_\mu^4 G_\nu^2 G^{7\mu} G^{3\nu} + \frac{\sqrt{3}g_3^2}{16} G_\mu^4 G_\nu^2 G^{7\mu} G^{8\nu} \\
& -\frac{\sqrt{3}g_3^2}{16} G_\mu^4 G_\nu^2 G^{8\mu} G^{7\nu} + \frac{g_3^2}{16} G_\mu^4 G_\nu^3 G^{1\mu} G^{6\nu} + -\frac{g_3^2}{16} G_\mu^4 G_\nu^3 G^{2\mu} G^{7\nu} \\
& \frac{g_3^2}{16} G_\mu^4 G_\nu^3 G^{3\mu} G^{4\nu} + -\frac{g_3^2}{16} G_\mu^4 G_\nu^3 G^{4\mu} G^{3\nu} + -\frac{\sqrt{3}g_3^2}{16} G_\mu^4 G_\nu^3 G^{4\mu} G^{8\nu} \\
& -\frac{g_3^2}{16} G_\mu^4 G_\nu^3 G^{6\mu} G^{1\nu} + \frac{g_3^2}{16} G_\mu^4 G_\nu^3 G^{7\mu} G^{2\nu} + \frac{\sqrt{3}g_3^2}{16} G_\mu^4 G_\nu^3 G^{8\mu} G^{4\nu} \\
& -\frac{g_3^2}{8} G_\mu^4 G_\nu^5 G^{1\mu} G^{2\nu} + \frac{g_3^2}{8} G_\mu^4 G_\nu^5 G^{2\mu} G^{1\nu} + -\frac{g_3^2}{16} G_\mu^4 G_\nu^5 G^{4\mu} G^{5\nu} \\
& \frac{g_3^2}{16} G_\mu^4 G_\nu^5 G^{5\mu} G^{4\nu} + \frac{g_3^2}{16} G_\mu^4 G_\nu^5 G^{6\mu} G^{7\nu} + -\frac{g_3^2}{16} G_\mu^4 G_\nu^5 G^{7\mu} G^{6\nu} \\
& -\frac{3g_3^2}{16} G_\mu^4 G_\nu^5 G^{4\mu} G^{5\nu} + \frac{3g_3^2}{16} G_\mu^4 G_\nu^5 G^{5\mu} G^{4\nu} + -\frac{3g_3^2}{16} G_\mu^4 G_\nu^5 G^{6\mu} G^{7\nu} \\
& \frac{3g_3^2}{16} G_\mu^4 G_\nu^5 G^{7\mu} G^{6\nu} + \frac{g_3^2}{8} G_\mu^4 G_\nu^6 G^{1\mu} G^{3\nu} + -\frac{g_3^2}{8} G_\mu^4 G_\nu^6 G^{3\mu} G^{1\nu} \\
& -\frac{g_3^2}{16} G_\mu^4 G_\nu^6 G^{4\mu} G^{6\nu} + -\frac{g_3^2}{16} G_\mu^4 G_\nu^6 G^{5\mu} G^{7\nu} + \frac{g_3^2}{16} G_\mu^4 G_\nu^6 G^{6\mu} G^{4\nu} \\
& \frac{g_3^2}{16} G_\mu^4 G_\nu^6 G^{7\mu} G^{5\nu} + -\frac{g_3^2}{8} G_\mu^4 G_\nu^7 G^{2\mu} G^{3\nu} + \frac{g_3^2}{8} G_\mu^4 G_\nu^7 G^{3\mu} G^{2\nu} \\
& -\frac{g_3^2}{16} G_\mu^4 G_\nu^7 G^{4\mu} G^{7\nu} + \frac{g_3^2}{16} G_\mu^4 G_\nu^7 G^{5\mu} G^{6\nu} + -\frac{g_3^2}{16} G_\mu^4 G_\nu^7 G^{6\mu} G^{5\nu} \\
& \frac{g_3^2}{16} G_\mu^4 G_\nu^7 G^{7\mu} G^{4\nu} + \frac{\sqrt{3}g_3^2}{16} G_\mu^4 G_\nu^8 G^{1\mu} G^{6\nu} + -\frac{\sqrt{3}g_3^2}{16} G_\mu^4 G_\nu^8 G^{2\mu} G^{7\nu}
\end{aligned}$$

$$\begin{aligned}
& \frac{\sqrt{3}g_3^2}{16} G_\mu^4 G_\nu^8 G^{3\mu} G^{4\nu} + -\frac{\sqrt{3}g_3^2}{16} G_\mu^4 G_\nu^8 G^{4\mu} G^{3\nu} + -\frac{3g_3^2}{16} G_\mu^4 G_\nu^8 G^{4\mu} G^{8\nu} \\
& -\frac{\sqrt{3}g_3^2}{16} G_\mu^4 G_\nu^8 G^{6\mu} G^{1\nu} + \frac{\sqrt{3}g_3^2}{16} G_\mu^4 G_\nu^8 G^{7\mu} G^{2\nu} + \frac{3g_3^2}{16} G_\mu^4 G_\nu^8 G^{8\mu} G^{4\nu} \\
& \frac{g_3^2}{16} G_\mu^5 G_\nu^1 G^{1\mu} G^{5\nu} + -\frac{g_3^2}{16} G_\mu^5 G_\nu^1 G^{2\mu} G^{4\nu} + -\frac{g_3^2}{16} G_\mu^5 G_\nu^1 G^{3\mu} G^{7\nu} \\
& \frac{g_3^2}{16} G_\mu^5 G_\nu^1 G^{4\mu} G^{2\nu} + -\frac{g_3^2}{16} G_\mu^5 G_\nu^1 G^{5\mu} G^{1\nu} + \frac{g_3^2}{16} G_\mu^5 G_\nu^1 G^{7\mu} G^{3\nu} \\
& -\frac{\sqrt{3}g_3^2}{16} G_\mu^5 G_\nu^1 G^{7\mu} G^{8\nu} + \frac{\sqrt{3}g_3^2}{16} G_\mu^5 G_\nu^1 G^{8\mu} G^{7\nu} + \frac{g_3^2}{16} G_\mu^5 G_\nu^2 G^{1\mu} G^{4\nu} \\
& \frac{g_3^2}{16} G_\mu^5 G_\nu^2 G^{2\mu} G^{5\nu} + -\frac{g_3^2}{16} G_\mu^5 G_\nu^2 G^{3\mu} G^{6\nu} + -\frac{g_3^2}{16} G_\mu^5 G_\nu^2 G^{4\mu} G^{1\nu} \\
& -\frac{g_3^2}{16} G_\mu^5 G_\nu^2 G^{5\mu} G^{2\nu} + \frac{g_3^2}{16} G_\mu^5 G_\nu^2 G^{6\mu} G^{3\nu} + -\frac{\sqrt{3}g_3^2}{16} G_\mu^5 G_\nu^2 G^{6\mu} G^{8\nu} \\
& \frac{\sqrt{3}g_3^2}{16} G_\mu^5 G_\nu^2 G^{8\mu} G^{6\nu} + \frac{g_3^2}{16} G_\mu^5 G_\nu^3 G^{1\mu} G^{7\nu} + \frac{g_3^2}{16} G_\mu^5 G_\nu^3 G^{2\mu} G^{6\nu} \\
& \frac{g_3^2}{16} G_\mu^5 G_\nu^3 G^{3\mu} G^{5\nu} + -\frac{g_3^2}{16} G_\mu^5 G_\nu^3 G^{5\mu} G^{3\nu} + -\frac{\sqrt{3}g_3^2}{16} G_\mu^5 G_\nu^3 G^{5\mu} G^{8\nu} \\
& -\frac{g_3^2}{16} G_\mu^5 G_\nu^3 G^{6\mu} G^{2\nu} + -\frac{g_3^2}{16} G_\mu^5 G_\nu^3 G^{7\mu} G^{1\nu} + \frac{\sqrt{3}g_3^2}{16} G_\mu^5 G_\nu^3 G^{8\mu} G^{5\nu} \\
& \frac{g_3^2}{8} G_\mu^5 G_\nu^4 G^{1\mu} G^{2\nu} + -\frac{g_3^2}{8} G_\mu^5 G_\nu^4 G^{2\mu} G^{1\nu} + \frac{g_3^2}{16} G_\mu^5 G_\nu^4 G^{4\mu} G^{5\nu} \\
& -\frac{g_3^2}{16} G_\mu^5 G_\nu^4 G^{5\mu} G^{4\nu} + -\frac{g_3^2}{16} G_\mu^5 G_\nu^4 G^{6\mu} G^{7\nu} + \frac{g_3^2}{16} G_\mu^5 G_\nu^4 G^{7\mu} G^{6\nu} \\
& \frac{3g_3^2}{16} G_\mu^5 G_\nu^4 G^{4\mu} G^{5\nu} + -\frac{3g_3^2}{16} G_\mu^5 G_\nu^4 G^{5\mu} G^{4\nu} + \frac{3g_3^2}{16} G_\mu^5 G_\nu^4 G^{6\mu} G^{7\nu} \\
& -\frac{3g_3^2}{16} G_\mu^5 G_\nu^4 G^{7\mu} G^{6\nu} + \frac{g_3^2}{8} G_\mu^5 G_\nu^6 G^{2\mu} G^{3\nu} + -\frac{g_3^2}{8} G_\mu^5 G_\nu^6 G^{3\mu} G^{2\nu} \\
& \frac{g_3^2}{16} G_\mu^5 G_\nu^6 G^{4\mu} G^{7\nu} + -\frac{g_3^2}{16} G_\mu^5 G_\nu^6 G^{5\mu} G^{6\nu} + \frac{g_3^2}{16} G_\mu^5 G_\nu^6 G^{6\mu} G^{5\nu} \\
& -\frac{g_3^2}{16} G_\mu^5 G_\nu^6 G^{7\mu} G^{4\nu} + \frac{g_3^2}{8} G_\mu^5 G_\nu^7 G^{1\mu} G^{3\nu} + -\frac{g_3^2}{8} G_\mu^5 G_\nu^7 G^{3\mu} G^{1\nu} \\
& -\frac{g_3^2}{16} G_\mu^5 G_\nu^7 G^{4\mu} G^{6\nu} + -\frac{g_3^2}{16} G_\mu^5 G_\nu^7 G^{5\mu} G^{7\nu} + \frac{g_3^2}{16} G_\mu^5 G_\nu^7 G^{6\mu} G^{4\nu}
\end{aligned}$$

$$\begin{aligned}
& \frac{g_3^2}{16} G_\mu^5 G_\nu^7 G^{7\mu} G^{5\nu} + \frac{\sqrt{3}g_3^2}{16} G_\mu^5 G_\nu^8 G^{1\mu} G^{7\nu} + \frac{\sqrt{3}g_3^2}{16} G_\mu^5 G_\nu^8 G^{2\mu} G^{6\nu} \\
& \frac{\sqrt{3}g_3^2}{16} G_\mu^5 G_\nu^8 G^{3\mu} G^{5\nu} + -\frac{\sqrt{3}g_3^2}{16} G_\mu^5 G_\nu^8 G^{5\mu} G^{3\nu} + -\frac{3g_3^2}{16} G_\mu^5 G_\nu^8 G^{5\mu} G^{8\nu} \\
& -\frac{\sqrt{3}g_3^2}{16} G_\mu^5 G_\nu^8 G^{6\mu} G^{2\nu} + -\frac{\sqrt{3}g_3^2}{16} G_\mu^5 G_\nu^8 G^{7\mu} G^{1\nu} + \frac{3g_3^2}{16} G_\mu^5 G_\nu^8 G^{8\mu} G^{5\nu} \\
& \frac{g_3^2}{16} G_\mu^6 G_\nu^1 G^{1\mu} G^{6\nu} + -\frac{g_3^2}{16} G_\mu^6 G_\nu^1 G^{2\mu} G^{7\nu} + \frac{g_3^2}{16} G_\mu^6 G_\nu^1 G^{3\mu} G^{4\nu} \\
& -\frac{g_3^2}{16} G_\mu^6 G_\nu^1 G^{4\mu} G^{3\nu} + -\frac{\sqrt{3}g_3^2}{16} G_\mu^6 G_\nu^1 G^{4\mu} G^{8\nu} + -\frac{g_3^2}{16} G_\mu^6 G_\nu^1 G^{6\mu} G^{1\nu} \\
& \frac{g_3^2}{16} G_\mu^6 G_\nu^1 G^{7\mu} G^{2\nu} + \frac{\sqrt{3}g_3^2}{16} G_\mu^6 G_\nu^1 G^{8\mu} G^{4\nu} + \frac{g_3^2}{16} G_\mu^6 G_\nu^2 G^{1\mu} G^{7\nu} \\
& \frac{g_3^2}{16} G_\mu^6 G_\nu^2 G^{2\mu} G^{6\nu} + \frac{g_3^2}{16} G_\mu^6 G_\nu^2 G^{3\mu} G^{5\nu} + -\frac{g_3^2}{16} G_\mu^6 G_\nu^2 G^{5\mu} G^{3\nu} \\
& -\frac{\sqrt{3}g_3^2}{16} G_\mu^6 G_\nu^2 G^{5\mu} G^{8\nu} + -\frac{g_3^2}{16} G_\mu^6 G_\nu^2 G^{6\mu} G^{2\nu} + -\frac{g_3^2}{16} G_\mu^6 G_\nu^2 G^{7\mu} G^{1\nu} \\
& \frac{\sqrt{3}g_3^2}{16} G_\mu^6 G_\nu^2 G^{8\mu} G^{5\nu} + -\frac{g_3^2}{16} G_\mu^6 G_\nu^3 G^{1\mu} G^{4\nu} + -\frac{g_3^2}{16} G_\mu^6 G_\nu^3 G^{2\mu} G^{5\nu} \\
& \frac{g_3^2}{16} G_\mu^6 G_\nu^3 G^{3\mu} G^{6\nu} + \frac{g_3^2}{16} G_\mu^6 G_\nu^3 G^{4\mu} G^{1\nu} + \frac{g_3^2}{16} G_\mu^6 G_\nu^3 G^{5\mu} G^{2\nu} \\
& -\frac{g_3^2}{16} G_\mu^6 G_\nu^3 G^{6\mu} G^{3\nu} + \frac{\sqrt{3}g_3^2}{16} G_\mu^6 G_\nu^3 G^{6\mu} G^{8\nu} + -\frac{\sqrt{3}g_3^2}{16} G_\mu^6 G_\nu^3 G^{8\mu} G^{6\nu} \\
& -\frac{g_3^2}{8} G_\mu^6 G_\nu^4 G^{1\mu} G^{3\nu} + \frac{g_3^2}{8} G_\mu^6 G_\nu^4 G^{3\mu} G^{1\nu} + \frac{g_3^2}{16} G_\mu^6 G_\nu^4 G^{4\mu} G^{6\nu} \\
& \frac{g_3^2}{16} G_\mu^6 G_\nu^4 G^{5\mu} G^{7\nu} + -\frac{g_3^2}{16} G_\mu^6 G_\nu^4 G^{6\mu} G^{4\nu} + -\frac{g_3^2}{16} G_\mu^6 G_\nu^4 G^{7\mu} G^{5\nu} \\
& -\frac{g_3^2}{8} G_\mu^6 G_\nu^5 G^{2\mu} G^{3\nu} + \frac{g_3^2}{8} G_\mu^6 G_\nu^5 G^{3\mu} G^{2\nu} + -\frac{g_3^2}{16} G_\mu^6 G_\nu^5 G^{4\mu} G^{7\nu} \\
& \frac{g_3^2}{16} G_\mu^6 G_\nu^5 G^{5\mu} G^{6\nu} + -\frac{g_3^2}{16} G_\mu^6 G_\nu^5 G^{6\mu} G^{5\nu} + \frac{g_3^2}{16} G_\mu^6 G_\nu^5 G^{7\mu} G^{4\nu} \\
& \frac{g_3^2}{8} G_\mu^6 G_\nu^7 G^{1\mu} G^{2\nu} + -\frac{g_3^2}{8} G_\mu^6 G_\nu^7 G^{2\mu} G^{1\nu} + \frac{g_3^2}{16} G_\mu^6 G_\nu^7 G^{4\mu} G^{5\nu} \\
& -\frac{g_3^2}{16} G_\mu^6 G_\nu^7 G^{5\mu} G^{4\nu} + -\frac{g_3^2}{16} G_\mu^6 G_\nu^7 G^{6\mu} G^{7\nu} + \frac{g_3^2}{16} G_\mu^6 G_\nu^7 G^{7\mu} G^{6\nu}
\end{aligned}$$

$$\begin{aligned}
& -\frac{3g_3^2}{16} G_\mu^6 G_\nu^7 G^{4\mu} G^{5\nu} + \frac{3g_3^2}{16} G_\mu^6 G_\nu^7 G^{5\mu} G^{4\nu} + -\frac{3g_3^2}{16} G_\mu^6 G_\nu^7 G^{6\mu} G^{7\nu} \\
& \frac{3g_3^2}{16} G_\mu^6 G_\nu^7 G^{7\mu} G^{6\nu} + \frac{\sqrt{3}g_3^2}{16} G_\mu^6 G_\nu^8 G^{1\mu} G^{4\nu} + \frac{\sqrt{3}g_3^2}{16} G_\mu^6 G_\nu^8 G^{2\mu} G^{5\nu} \\
& -\frac{\sqrt{3}g_3^2}{16} G_\mu^6 G_\nu^8 G^{3\mu} G^{6\nu} + -\frac{\sqrt{3}g_3^2}{16} G_\mu^6 G_\nu^8 G^{4\mu} G^{1\nu} + -\frac{\sqrt{3}g_3^2}{16} G_\mu^6 G_\nu^8 G^{5\mu} G^{2\nu} \\
& \frac{\sqrt{3}g_3^2}{16} G_\mu^6 G_\nu^8 G^{6\mu} G^{3\nu} + -\frac{3g_3^2}{16} G_\mu^6 G_\nu^8 G^{6\mu} G^{8\nu} + \frac{3g_3^2}{16} G_\mu^6 G_\nu^8 G^{8\mu} G^{6\nu} \\
& \frac{g_3^2}{16} G_\mu^7 G_\nu^1 G^{1\mu} G^{7\nu} + \frac{g_3^2}{16} G_\mu^7 G_\nu^1 G^{2\mu} G^{6\nu} + \frac{g_3^2}{16} G_\mu^7 G_\nu^1 G^{3\mu} G^{5\nu} \\
& -\frac{g_3^2}{16} G_\mu^7 G_\nu^1 G^{5\mu} G^{3\nu} + -\frac{\sqrt{3}g_3^2}{16} G_\mu^7 G_\nu^1 G^{5\mu} G^{8\nu} + -\frac{g_3^2}{16} G_\mu^7 G_\nu^1 G^{6\mu} G^{2\nu} \\
& -\frac{g_3^2}{16} G_\mu^7 G_\nu^1 G^{7\mu} G^{1\nu} + \frac{\sqrt{3}g_3^2}{16} G_\mu^7 G_\nu^1 G^{8\mu} G^{5\nu} + -\frac{g_3^2}{16} G_\mu^7 G_\nu^2 G^{1\mu} G^{6\nu} \\
& \frac{g_3^2}{16} G_\mu^7 G_\nu^2 G^{2\mu} G^{7\nu} + -\frac{g_3^2}{16} G_\mu^7 G_\nu^2 G^{3\mu} G^{4\nu} + \frac{g_3^2}{16} G_\mu^7 G_\nu^2 G^{4\mu} G^{3\nu} \\
& \frac{\sqrt{3}g_3^2}{16} G_\mu^7 G_\nu^2 G^{4\mu} G^{8\nu} + \frac{g_3^2}{16} G_\mu^7 G_\nu^2 G^{6\mu} G^{1\nu} + -\frac{g_3^2}{16} G_\mu^7 G_\nu^2 G^{7\mu} G^{2\nu} \\
& -\frac{\sqrt{3}g_3^2}{16} G_\mu^7 G_\nu^2 G^{8\mu} G^{4\nu} + -\frac{g_3^2}{16} G_\mu^7 G_\nu^3 G^{1\mu} G^{5\nu} + \frac{g_3^2}{16} G_\mu^7 G_\nu^3 G^{2\mu} G^{4\nu} \\
& \frac{g_3^2}{16} G_\mu^7 G_\nu^3 G^{3\mu} G^{7\nu} + -\frac{g_3^2}{16} G_\mu^7 G_\nu^3 G^{4\mu} G^{2\nu} + \frac{g_3^2}{16} G_\mu^7 G_\nu^3 G^{5\mu} G^{1\nu} \\
& -\frac{g_3^2}{16} G_\mu^7 G_\nu^3 G^{7\mu} G^{3\nu} + \frac{\sqrt{3}g_3^2}{16} G_\mu^7 G_\nu^3 G^{7\mu} G^{8\nu} + -\frac{\sqrt{3}g_3^2}{16} G_\mu^7 G_\nu^3 G^{8\mu} G^{7\nu} \\
& \frac{g_3^2}{8} G_\mu^7 G_\nu^4 G^{2\mu} G^{3\nu} + -\frac{g_3^2}{8} G_\mu^7 G_\nu^4 G^{3\mu} G^{2\nu} + \frac{g_3^2}{16} G_\mu^7 G_\nu^4 G^{4\mu} G^{7\nu} \\
& -\frac{g_3^2}{16} G_\mu^7 G_\nu^4 G^{5\mu} G^{6\nu} + \frac{g_3^2}{16} G_\mu^7 G_\nu^4 G^{6\mu} G^{5\nu} + -\frac{g_3^2}{16} G_\mu^7 G_\nu^4 G^{7\mu} G^{4\nu} \\
& -\frac{g_3^2}{8} G_\mu^7 G_\nu^5 G^{1\mu} G^{3\nu} + \frac{g_3^2}{8} G_\mu^7 G_\nu^5 G^{3\mu} G^{1\nu} + \frac{g_3^2}{16} G_\mu^7 G_\nu^5 G^{4\mu} G^{6\nu} \\
& \frac{g_3^2}{16} G_\mu^7 G_\nu^5 G^{5\mu} G^{7\nu} + -\frac{g_3^2}{16} G_\mu^7 G_\nu^5 G^{6\mu} G^{4\nu} + -\frac{g_3^2}{16} G_\mu^7 G_\nu^5 G^{7\mu} G^{5\nu} \\
& -\frac{g_3^2}{8} G_\mu^7 G_\nu^6 G^{1\mu} G^{2\nu} + \frac{g_3^2}{8} G_\mu^7 G_\nu^6 G^{2\mu} G^{1\nu} + -\frac{g_3^2}{16} G_\mu^7 G_\nu^6 G^{4\mu} G^{5\nu}
\end{aligned}$$

$$\begin{aligned}
& \frac{g_3^2}{16} G_\mu^7 G_\nu^6 G^{5\mu} G^{4\nu} + \frac{g_3^2}{16} G_\mu^7 G_\nu^6 G^{6\mu} G^{7\nu} + -\frac{g_3^2}{16} G_\mu^7 G_\nu^6 G^{7\mu} G^{6\nu} \\
& \frac{3g_3^2}{16} G_\mu^7 G_\nu^6 G^{4\mu} G^{5\nu} + -\frac{3g_3^2}{16} G_\mu^7 G_\nu^6 G^{5\mu} G^{4\nu} + \frac{3g_3^2}{16} G_\mu^7 G_\nu^6 G^{6\mu} G^{7\nu} \\
& -\frac{3g_3^2}{16} G_\mu^7 G_\nu^6 G^{7\mu} G^{6\nu} + \frac{\sqrt{3}g_3^2}{16} G_\mu^7 G_\nu^8 G^{1\mu} G^{5\nu} + -\frac{\sqrt{3}g_3^2}{16} G_\mu^7 G_\nu^8 G^{2\mu} G^{4\nu} \\
& -\frac{\sqrt{3}g_3^2}{16} G_\mu^7 G_\nu^8 G^{3\mu} G^{7\nu} + \frac{\sqrt{3}g_3^2}{16} G_\mu^7 G_\nu^8 G^{4\mu} G^{2\nu} + -\frac{\sqrt{3}g_3^2}{16} G_\mu^7 G_\nu^8 G^{5\mu} G^{1\nu} \\
& \frac{\sqrt{3}g_3^2}{16} G_\mu^7 G_\nu^8 G^{7\mu} G^{3\nu} + -\frac{3g_3^2}{16} G_\mu^7 G_\nu^8 G^{7\mu} G^{8\nu} + \frac{3g_3^2}{16} G_\mu^7 G_\nu^8 G^{8\mu} G^{7\nu} \\
& -\frac{\sqrt{3}g_3^2}{16} G_\mu^8 G_\nu^4 G^{1\mu} G^{6\nu} + \frac{\sqrt{3}g_3^2}{16} G_\mu^8 G_\nu^4 G^{2\mu} G^{7\nu} + -\frac{\sqrt{3}g_3^2}{16} G_\mu^8 G_\nu^4 G^{3\mu} G^{4\nu} \\
& \frac{\sqrt{3}g_3^2}{16} G_\mu^8 G_\nu^4 G^{4\mu} G^{3\nu} + \frac{3g_3^2}{16} G_\mu^8 G_\nu^4 G^{4\mu} G^{8\nu} + \frac{\sqrt{3}g_3^2}{16} G_\mu^8 G_\nu^4 G^{6\mu} G^{1\nu} \\
& -\frac{\sqrt{3}g_3^2}{16} G_\mu^8 G_\nu^4 G^{7\mu} G^{2\nu} + -\frac{3g_3^2}{16} G_\mu^8 G_\nu^4 G^{8\mu} G^{4\nu} + -\frac{\sqrt{3}g_3^2}{16} G_\mu^8 G_\nu^5 G^{1\mu} G^{7\nu} \\
& -\frac{\sqrt{3}g_3^2}{16} G_\mu^8 G_\nu^5 G^{2\mu} G^{6\nu} + -\frac{\sqrt{3}g_3^2}{16} G_\mu^8 G_\nu^5 G^{3\mu} G^{5\nu} + \frac{\sqrt{3}g_3^2}{16} G_\mu^8 G_\nu^5 G^{5\mu} G^{3\nu} \\
& \frac{3g_3^2}{16} G_\mu^8 G_\nu^5 G^{5\mu} G^{8\nu} + \frac{\sqrt{3}g_3^2}{16} G_\mu^8 G_\nu^5 G^{6\mu} G^{2\nu} + \frac{\sqrt{3}g_3^2}{16} G_\mu^8 G_\nu^5 G^{7\mu} G^{1\nu} \\
& -\frac{3g_3^2}{16} G_\mu^8 G_\nu^5 G^{8\mu} G^{5\nu} + -\frac{\sqrt{3}g_3^2}{16} G_\mu^8 G_\nu^6 G^{1\mu} G^{4\nu} + -\frac{\sqrt{3}g_3^2}{16} G_\mu^8 G_\nu^6 G^{2\mu} G^{5\nu} \\
& \frac{\sqrt{3}g_3^2}{16} G_\mu^8 G_\nu^6 G^{3\mu} G^{6\nu} + \frac{\sqrt{3}g_3^2}{16} G_\mu^8 G_\nu^6 G^{4\mu} G^{1\nu} + \frac{\sqrt{3}g_3^2}{16} G_\mu^8 G_\nu^6 G^{5\mu} G^{2\nu} \\
& -\frac{\sqrt{3}g_3^2}{16} G_\mu^8 G_\nu^6 G^{6\mu} G^{3\nu} + \frac{3g_3^2}{16} G_\mu^8 G_\nu^6 G^{6\mu} G^{8\nu} + -\frac{3g_3^2}{16} G_\mu^8 G_\nu^6 G^{8\mu} G^{6\nu} \\
& -\frac{\sqrt{3}g_3^2}{16} G_\mu^8 G_\nu^7 G^{1\mu} G^{5\nu} + \frac{\sqrt{3}g_3^2}{16} G_\mu^8 G_\nu^7 G^{2\mu} G^{4\nu} + \frac{\sqrt{3}g_3^2}{16} G_\mu^8 G_\nu^7 G^{3\mu} G^{7\nu} \\
& -\frac{\sqrt{3}g_3^2}{16} G_\mu^8 G_\nu^7 G^{4\mu} G^{2\nu} + \frac{\sqrt{3}g_3^2}{16} G_\mu^8 G_\nu^7 G^{5\mu} G^{1\nu} + -\frac{\sqrt{3}g_3^2}{16} G_\mu^8 G_\nu^7 G^{7\mu} G^{3\nu} \\
& \frac{3g_3^2}{16} G_\mu^8 G_\nu^7 G^{7\mu} G^{8\nu} + -\frac{3g_3^2}{16} G_\mu^8 G_\nu^7 G^{8\mu} G^{7\nu}
\end{aligned}$$

4 EWSB Mass Matrices

4.1 VEV Substitutions

$$H^0 \rightarrow H^0 + \frac{\sqrt{2}v}{2}$$

$$H^{0\dagger} \rightarrow H^{0\dagger} + \frac{\sqrt{2}v}{2}$$

4.2 Stationary Condition

$$\begin{aligned} \frac{\partial V}{\partial v} &= v (\lambda_2 v^2 + m_1^2) \\ &= 0 \end{aligned}$$

4.3 Neutral Scalar Mass Matrix

Neutral basis

$$b_1 = \text{Re}(H^0)$$

$$b_2 = \text{Im}(H^0)$$

Neutral scalar mass-matrix entries

$$M_{\text{neutral},11}^2 = 3\lambda_2 v^2 + m_1^2$$

$$M_{\text{neutral},12}^2 = 0$$

$$M_{\text{neutral},21}^2 = 0$$

$$M_{\text{neutral},22}^2 = \lambda_2 v^2 + m_1^2$$

4.4 Scalar Hermitian Blocks

$$M_1^2[(H^+)^\dagger; H^+] = [\lambda_2 v^2 + m_1^2]$$

4.5 Scalar Holomorphic Blocks

No blocks generated.

4.6 Fermion Blocks

$$M_1[d^{C+1/3,\bar{b}};q_b^{-1/3}] = \left[\frac{\sqrt{2}Y_1v}{2} \right]$$

$$M_2[d^{C+1/3,\bar{g}};q_g^{-1/3}] = \left[\frac{\sqrt{2}Y_1v}{2} \right]$$

$$M_3[d^{C+1/3,\bar{r}};q_r^{-1/3}] = \left[\frac{\sqrt{2}Y_1v}{2} \right]$$

$$M_4[e^{C+};l^-] = \left[\frac{\sqrt{2}Y_3v}{2} \right]$$

$$M_5[u^{C-2/3,\bar{b}};q_b^{+2/3}] = \left[\frac{\sqrt{2}Y_2v}{2} \right]$$

$$M_6[u^{C-2/3,\bar{g}};q_g^{+2/3}] = \left[\frac{\sqrt{2}Y_2v}{2} \right]$$

$$M_7[u^{C-2/3,\bar{r}};q_r^{+2/3}] = \left[\frac{\sqrt{2}Y_2v}{2} \right]$$